

EDA_Optimising_NYC_Taxis_RAJESH WARAN R S

Include your visualisations, analysis, results, insights, and outcomes. Explain your methodology and approach to the tasks. Add your conclusions to the sections.

1. Data Preparation

1.1. Loading the dataset

- The dataset consists of **NYC Yellow Taxi trip records for the year 2023**, provided as 12 monthly Parquet files, covering January through December 2023.
- Since the complete dataset is very large, loading all 12 files simultaneously would require substantial memory. Therefore, the files were processed **one month at a time**.
- The dataset was loaded using Pandas, and the monthly files were subsequently combined after applying the sampling methodology described below.

1.1.1. Sample the data and combine the files

Given the size of the original dataset, a **5% random sample was taken from each pickup date and hour** rather than loading and analysing the entire dataset.

1. The resulting annual sampled dataset contains **1,896,400 records and 22 columns**.
2. The sampled dataset was saved as:
`nyc_taxi_2023_sampled.parquet`
3. This intermediate file was then reloaded for the **Data Cleaning** stage.

2. Data Cleaning

2.1. Fixing Columns

2.1.1. Fix the index

Because we combined many monthly/hourly samples, the original indexes are no longer meaningful. Hence, drop the messy index and create a clean sequential index.

2.1.2. Combine the two airport_fee columns

Two columns, Airport_fee and airport_fee, were identified in the dataset, likely resulting from inconsistent column naming across the monthly source files. The two columns were consolidated into a single airport_fee column. Existing values were retained from Airport_fee, while missing values were supplemented using values from airport_fee. The redundant Airport_fee column was then removed.

Outcome: The dataset now contains a single standardized airport_fee variable without losing available fee information.

2.1.3 Fix columns with negative monetary values

Negative values were checked across the numeric columns of the sampled dataset. There are no negative values for columns (fare_amount , RatecodeID).

Negative observations were identified in extra, mta_tax, improvement_surcharge, total_amount, congestion_surcharge, and airport_fee.

The affected monetary columns were cleaned using a lower bound of zero with Panda's clip(lower=0). This replaced negative monetary values with zero while leaving positive values, zeros and missing values unchanged.

A post-cleaning validation was performed, which returned an empty dictionary {}, confirming that **no negative numeric values remained in the dataset.**

2.2. Handling Missing Values

2.2.1. Find the proportion of missing values in each column

The missing-value analysis identified 64,874 records (3.42% of the sampled dataset) with missing values across five fields: passenger_count, airport_fee, congestion_surcharge, store_and_fwd_flag, and RatecodeID. The missing values were found to occur in the same set of records. These findings were used to determine the appropriate treatment for each affected variable in the subsequent data-cleaning steps.

2.2.2. Handling missing values in passenger_count

The `passenger_count` column initially contained 64,874 missing values and 29,681 zero values. The median passenger count was 1.0. Missing values were imputed with the median, and zero passenger counts were also replaced with 1.0 after inspection showed that these records represented otherwise valid trips with positive distance and fare values. Post-cleaning, no zero or missing passenger counts remained.

2.2.3. Handle missing values in RatecodeID

`RatecodeID` had 64,874 missing observations (3.42%). Since `RatecodeID` represents a categorical code, mode imputation was used rather than mean or median imputation. `RatecodeID = 1` was the most frequently occurring category, so all missing values were replaced with 1.0. Post-imputation validation confirmed that no missing values remained.

2.2.4. Impute NaN in congestion_surcharge

`congestion_surcharge` is a monetary numeric variable. Its distribution is highly concentrated at 2.5, while a substantial number of records have 0.0. The mean (2.31) is therefore lower than the median (2.5). The median was selected for imputation because it better represents the typical non-missing surcharge without being influenced by the zero-valued observations.

2.2.5. Impute NaN in store_and_fwd_flag

`store_and_fwd_flag` contained 64,874 missing observations (3.42%). The variable is categorical, with `N` being the dominant category. Therefore, missing values were imputed using the mode (`N`). After imputation, no missing values remained in the column.

2.3. Handling Outliers and Standardising Values

2.3.1. Check outliers in payment type, trip distance and tip amount columns

1) 35 records were identified with trip distance below 0.1 miles and fare amount above \$300. These observations showed an unusual combination

of extremely short/zero recorded distance and exceptionally high fares, indicating potential trip-registration errors. These records were therefore flagged for outlier treatment. And removed as part of the treatment.

- 2.3.2.** 63 records were identified where the pickup and drop-off zones differed, but both trip distance and fare amount were recorded as zero. Since trips between different zones would normally be expected to have a non-zero distance and fare, these records were flagged as potential trip-registration errors. And removed as part of the treatment.
- 2.3.3.** 46 records had trip distances exceeding 250 miles. Several observations showed extreme distances that were inconsistent with the recorded trip duration and fare amount, indicating erroneous trip registration. These records were flagged for outlier treatment. And removed as part of the treatment.
- 2.3.4.** Invalid payment types: 64,874 records had `payment_type = 0`, which is not defined as a valid payment category in the data dictionary. Since the actual payment type cannot be reliably inferred, these records were removed from the analysis, which is an invalid/unrecognized category, and there is no reliable basis to infer the actual payment type, since this is only about 3.42% of your 1,896,400 records.
- 2.3.5.** Trips with fare amounts above 300 and trip distances below 20 miles were identified as extreme fare–distance inconsistencies. Eight such records were removed as likely recording errors rather than genuine trips.
- 2.3.6.** Removed 7+ passenger counts as there are very few instances.
- 2.3.7.** Outlier Handling: Potential outliers were identified using domain-based checks. Records with passenger counts above 6, trip distances above 250 miles, invalid payment type 0, zero-distance/zero-fare trips between different zones, and extreme fare amounts relative to short trip distances were removed. Final validation confirmed zero remaining records for these identified outlier conditions.

3. Exploratory Data Analysis

3.1. General EDA: Finding Patterns and Trends

3.1.1. Classify variables into categorical and numerical

VendorID: Categorical

tpep_pickup_datetime: Categorical

tpep_dropoff_datetime: Categorical

passenger_count: Numerical

trip_distance: Numerical

RatecodeID: Categorical

PULocationID: Categorical

DOLocationID: Categorical

payment_type: Categorical

pickup_hour: Categorical

Trip_duration: Numerical

***The below monetary parameters are all : Numerical ***

- fare_amount
- extra
- mta_tax
- tip_amount
- tolls_amount
- improvement_surcharge
- total_amount
- congestion_surcharge
- airport_fee

3.1.2. Analyse the distribution of taxi pickups by hours, days of the week, and months

Taxi pickups by hours:

Taxi demand is lowest during the early morning hours, particularly around 4–5 AM. Demand increases from the morning onwards and peaks around 6 PM, indicating strong evening demand. This suggests that taxi availability and driver capacity may need to be higher during afternoon and evening hours.

Taxi pickups by days of the week:

The sampled data shows higher taxi pickup volume on Thursday compared with the other days.

Taxi pickups by months:

May month records the highest number of taxi pickups at approximately 167.1K, making it the peak month in the sampled 2023 data. Pickup volume then declines during June–September, reaching the lowest level in September at approximately 132.3K, before increasing again in October.

3.1.3. Filter out the zero/negative values in fares, distance and tips

Zero/negative values in fare, tip, total amount and trip distance were filtered out to create a non-zero dataset for the subsequent financial analysis. This resulted in 1,398,405 records being retained from the cleaned dataset.

3.1.4. Analyse the monthly revenue trends

Monthly revenue varies across the year. May and October record the highest revenue at approximately ₹3.8 million each, while February and August record the lowest at around ₹2.9 million. Revenue recovers from September, reaching its annual peak in October.

3.1.5. Find the proportion of each quarter's revenue in the yearly revenue

Quarterly revenue is relatively evenly distributed across the year. Q4 contributes the highest share of annual revenue at 27.2%, followed by Q2 at 26.8%. Q3 has the lowest contribution at 22.2%, while Q1 contributes 23.9%

3.1.6. Analyse and visualise the relationship between distance and fare amount

Trip distance and fare amount show a very strong positive relationship, with a correlation of 0.9514. This indicates that longer trips generally result in higher fares. The scatter plot also shows some variability and a

few unusually high fare values, indicating that factors other than distance may also influence the fare amount.

- 3.1.7.** Longer duration is associated with higher fare to some extent, but the relationship is much weaker than the relationship between distance and fare. Trip duration and fare amount show a positive but moderate relationship, with a correlation of 0.3194.

- 3.1.8. Analyse the relationship between fare/tips and trips/passengers**
Passenger count has a negligible linear relationship with fare amount, with a correlation of approximately 0.04. The scatter plot shows that fare values are widely distributed across all passenger-count categories, indicating that the number of passengers is not a major driver of the fare amount.

Trip distance and tip amount show a strong positive correlation of approximately 0.7951. This indicates that longer trips tend to be associated with higher tip amounts, although there is considerable variation in tipping behaviour across individual trips.

- 3.1.9. Analyse the distribution of different payment types**
Credit card is the predominant payment method, accounting for 81.5% of the sampled taxi trips, while cash accounts for 17.2%. No-charge and disputed trips represent only a small proportion of trips, at 0.5% and 0.7% respectively.

- 3.1.10. Load the taxi zones shapefile and display it**
Loaded the `taxi_zones.shp` file using GeoPandas and visualise the geographical distribution of NYC taxi zones.

- 3.1.11. Merge the zone data with trips data**
Merged taxi zone information with trip records using `PULocationID` from the trip data and `LocationID` from the zone dataset. This adds zone and borough information to each trip.

- 3.1.12. Find the number of trips for each zone/location ID**
Grouped the trip records by `PULocationID` and calculate the number of trips originating from each pickup location. Sort the results in descending order to identify the highest-demand pickup locations.

- 3.1.13. Add the number of trips for each zone to the zones dataframe**
The trip data was grouped by `PULocationID` to calculate the total number of pickup trips for each location. These trip counts were then merged with the taxi zones GeoDataFrame using `LocationID`.

This created a zone-level dataset containing the zone name, borough, geographical boundary, and total number of pickup trips, which was used for geographical analysis.

3.1.14. Plot a map of the zones showing number of trips

A color-coded choropleth map was created using the `trip_count` column. Darker and lighter shades represent different levels of taxi pickup activity across NYC zones.

3.1.15. Conclude with results

Taxi pickup activity is not evenly distributed across NYC zones.

This indicates that taxi demand is strongly concentrated around high-density business, commercial, tourist, and transportation hubs.

These high-demand zones can be important for driver positioning, fleet allocation, and demand forecasting.

Taxi operators can use zone-level demand patterns to position drivers closer to high-demand locations and improve vehicle availability during peak periods.

3.2. Detailed EDA: Insights and Strategies

3.2.1. Identify slow routes by comparing average speeds on different routes

Trip records were grouped by pickup zone, drop-off zone, and pickup hour. The average trip distance and average trip duration were calculated for each route-hour combination. Average speed was then calculated in miles per hour using the average distance and average duration. Routes with lower average speeds indicate potential traffic bottlenecks or slower-moving routes, particularly during specific hours of the day. This analysis helps identify route-hour combinations that may require closer investigation for operational inefficiencies.

3.2.2. Calculate the hourly number of trips and identify the busy hours

The sampled trip data was grouped by pickup hour to understand the variation in taxi demand throughout the day. The analysis showed that trip volumes increase from the morning and remain high through the afternoon and evening. The busiest hour was 18:00 (6 PM), with 128,061 sampled trips across the year. This indicates that the evening period represents a significant period of taxi demand and may require greater availability of taxis to handle customer demand.

3.2.3. Scale up the number of trips from above to find the actual number of trips

Since the analysis was performed using a 5% sample of the original taxi trip records, the observed trip counts were scaled using the sampling fraction of 0.05 to estimate the actual number of trips. For example, the busiest hour, 18:00, had 128,061 sampled trips. Scaling this by the sampling fraction gives an estimated 2,561,220 actual trips during the 18:00 hour across the year. The scaling approach is: Estimated actual trips = Sampled trips ÷ Sampling fraction.

3.2.4. Compare hourly traffic on weekdays and weekends

Weekday and weekend taxi demand show distinct hourly patterns. Weekday demand increases substantially from the morning and reaches its highest level around 18:00, indicating a strong evening peak. Weekend demand is lower overall and more evenly distributed across the daytime and evening hours. Both segments experience relatively low demand during the early-morning period. Identifying these busy and quiet periods can help optimize taxi availability, driver scheduling, vehicle positioning and utilization, while reducing idle capacity during low-demand hours.

3.2.5. Identify the top 10 zones with high hourly pickups and drops

Understanding these high-traffic zones can help taxi operators and fleet managers:

- Position taxis closer to areas with high expected demand.
- Plan driver availability during peak hours.
- Reduce passenger waiting time.
- Improve fleet utilization.
- Identify locations where pickup and drop-off demand differ significantly.

3.2.6. Find the ratio of pickups and dropoffs in each zone

This analysis helps identify zones that primarily generate trips versus zones that primarily attract trips, which can support taxi positioning and fleet planning.

3.2.7. Identify the top zones with high traffic during night hours
Night hours were defined as 11 PM to 5 AM (23:00–05:00). The data was first filtered for these hours, and the number of pickups and drop-offs was then calculated by zone.

3.2.8. The analysis identified the zones with the highest pickup and drop-off activity specifically during nighttime rather than using the overall top-performing zones. For the sampled data, there were 214,182 night-hour trips. The highest pickup activity was observed in Location IDs 79, 249, 132, 48 and 148, while Location IDs 79, 48, 170, 68 and 107 were among the leading drop-off locations. This helps identify areas requiring greater taxi availability during late-night and early-morning periods and can support driver allocation and demand planning.

3.2.9. Find the revenue share for nighttime and daytime hours
Night revenue share: 12.00% and Day revenue share: 88.00%
The three analyses together provide a view of where trips originate and terminate, where nighttime demand is concentrated, and how revenue is distributed across operating hours. These findings provide the foundation for the subsequent pricing-strategy analysis.

3.2.10. For the different passenger counts, find the average fare per mile per passenger
The analysis compares the average fare per mile across different passenger counts after adjusting the fare per mile for the number of passengers. This helps assess how the effective fare burden varies with changes in passenger occupancy. The results can be used to understand whether trips with different passenger counts show meaningful differences in fare efficiency.

3.2.11. Find the average fare per mile by hours of the day and by days of the week
The average fare per mile varies across both hours of the day and days of the week. At the hourly level, the highest average fare per mile is observed around 5 AM (14.56), while the lowest is around 9 PM (9.59). Across days, Sunday records the highest average fare per mile (12.65), while Friday records the lowest (10.98). These variations indicate that fare-per-mile patterns change depending on both time of day and day of week. Such patterns can be considered when evaluating time-based pricing strategies.

3.2.12. Analyse the average fare per mile for the different vendors
The analysis shows a clear difference between the two vendors across the hours of the day. Vendor 2 consistently records a higher average fare per mile than Vendor 1 across almost all hours. Vendor 2 also shows greater variation during the day, with its highest average fare per mile occurring around 5 AM (~17.7). Vendor 1 follows a comparatively stable pattern, generally ranging between approximately 6.5 and 9.0 per mile. Overall, the analysis indicates that fare-per-mile varies by both

vendor and pickup hour, which can be considered when evaluating vendor-specific and time-based pricing patterns.

3.2.13. Compare the fare rates of different vendors in a distance-tiered fashion

The analysis shows that average fare per mile decreases as trip distance increases for both vendors. For trips up to 2 miles, Vendor 2 has a substantially higher average fare per mile (17.93) compared with Vendor 1 (9.92). For trips between 2 and 5 miles, the rates are much closer, while for trips above 5 miles they are also very similar. Overall, the results suggest that short-distance trips have a higher fare per mile, whereas the effective fare per mile declines for longer trips. The difference between vendors is most pronounced for trips of up to 2 miles and becomes relatively small for longer trips.

3.2.14. Analyse the tip percentages

The analysis shows that tip percentage varies with trip distance, passenger count, and pickup time.

- Trip distance: Short trips have the highest average tip percentage. Trips up to 2 miles record an average tip percentage of 23.96%, compared with 18.76% for trips between 2–5 miles and 17.87% for trips above 5 miles. This suggests that the percentage of fare given as a tip tends to decrease as trip distance increases.
- Passenger count: The average tip percentage is highest for 1 passenger (22.02%) and generally decreases as passenger count increases, reaching the lowest level for 4 passengers (17.46%). It increases again for 5 and 6 passengers, so the relationship is not completely linear.
- Pickup time: Tip percentages also vary by hour. The highest average is observed at midnight (26.76%), followed by 3 PM (25.38%), while the lowest is around 5 AM (17.48%).

3.2.15. Analyse the trends in passenger count

Passenger demand is highest during the afternoon and evening, particularly around 5–6 PM, while early morning hours have substantially lower demand. Saturday also records the highest total passenger count and average occupancy. These patterns can help taxi operators align fleet availability with periods of higher passenger demand and plan resources more effectively.

3.2.16. Analyse the variation of passenger counts across zones

Passenger volumes are concentrated in major airport, business, entertainment, and transportation hubs. JFK Airport records the highest passenger count in the sampled data, while several high-volume Manhattan zones such as Midtown Center, Upper East Side, Times Square and Penn Station also show substantial passenger demand. This indicates that demand is geographically concentrated, which can help taxi

operators position vehicles and allocate fleet capacity around high-demand zones.

3.2.17. Analyse the pickup/dropoff zones or times when extra charges are applied more frequently.

Extra charges were applied to 62.14% of sampled trips. Their frequency was particularly high during the late afternoon and evening, with 18:00 recording the highest number of trips with an extra charge. Several high-traffic pickup zones, including Zone IDs 138, 161 and 237, also recorded high frequencies of extra-charge trips. The results indicate that both pickup location and time of day are associated with the frequency of extra charges.

4. Conclusions

4.1. Final Insights and Recommendations

4.1.1. Recommendations to optimize routing and dispatching based on demand patterns and operational inefficiencies.

****Key Insights****

Taxi demand rises substantially during the daytime and peaks around 18:00. Several routes show very low average speeds, indicating potential congestion or route inefficiencies.

Demand is concentrated in specific high-volume pickup zones.

Night-time demand remains significant, particularly in selected zones.

****Recommendations****

Increase taxi availability before and during the 17:00–19:00 peak period to match rising demand.

Pre-position vehicles near high-demand pickup zones such as Midtown, Upper East Side and major airports/stations before peak periods.

Use historical zone-hour demand patterns to dynamically reposition idle taxis rather than waiting for requests.

For consistently slow routes, consider alternative routing or dispatching vehicles from closer zones to reduce pickup and travel time.

Maintain targeted night-time availability in zones identified as having high overnight demand.

****Business objective****

Improve vehicle utilisation, reduce passenger waiting time and minimise supply-demand gaps.

4.1.2. Suggestions on strategically positioning cabs across different zones to make best use of insights uncovered by analysing trip trends across time, days and months.

****Key Insights****

1. Taxi activity is highly concentrated geographically rather than evenly distributed across all zones.
2. Manhattan's major commercial and transportation zones generate particularly high pickup volumes.
3. Airport zones such as JFK and LaGuardia are also among the important demand locations.
4. Demand patterns differ between weekdays, weekends and night hours.

****Recommendations****

Create zone-based demand clusters such as:

Manhattan commercial/business zones

Airport/transportation zones

Residential zones

Night-time demand zones

Position more taxis in high-demand zones during their specific peak hours, rather than keeping supply fixed throughout the day.

Use weekday and weekend patterns separately when planning vehicle deployment.

For airport and transportation zones, maintain sufficient availability around periods of consistently high demand.

During quiet periods, rebalance vehicles toward emerging/high-demand zones rather than allowing excess supply to remain in low-demand areas.

****Business objective****

Improve fleet utilisation while reducing idle time and increasing the probability of quickly matching passengers with available taxis.

4.1.3. Propose data-driven adjustments to the pricing strategy to maximize revenue while maintaining competitive rates with other vendors.

****Key Insights****

Average fare per mile varies considerably by hour and day.

In our analysis, average fare per mile was highest around 5 AM (14.56) and lowest around 21:00 (9.59).

Sunday had the highest average fare per mile among the days analysed, while Friday had the lowest.

Vendor-level analysis showed substantial differences in fare-per-mile patterns, particularly between the two vendors.

Fare per mile also changes significantly across distance tiers.

Tip percentages tend to be lower for longer-distance trips, while shorter trips showed higher average tip percentages.

Extra charges were present in 62.14% of sampled trips, with particularly high frequencies during late afternoon/evening hours.

****Recommendation****

Use time- and distance-based pricing analysis to identify opportunities for pricing optimisation rather than applying the same pricing approach across all periods.

Monitor fare-per-mile differences across vendors and distance tiers to maintain competitive and consistent pricing.

Consider differentiated pricing strategies for periods with strong demand, while ensuring that pricing remains competitive.

Review surcharge application by time and location, particularly during high-frequency periods, to ensure transparency and consistency.

Encourage customer retention by maintaining reasonable pricing on longer trips where tip percentages tend to decline.

****Business objective****

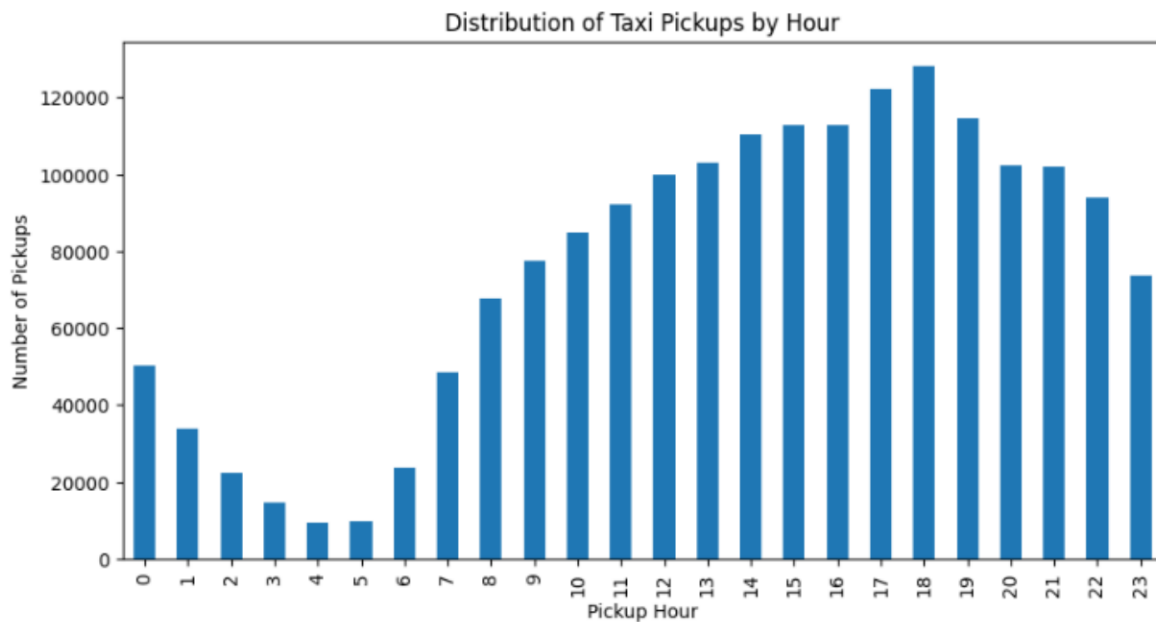
Improve revenue per trip while maintaining competitive fares and a positive customer experience.

****Final Concluding Statement****

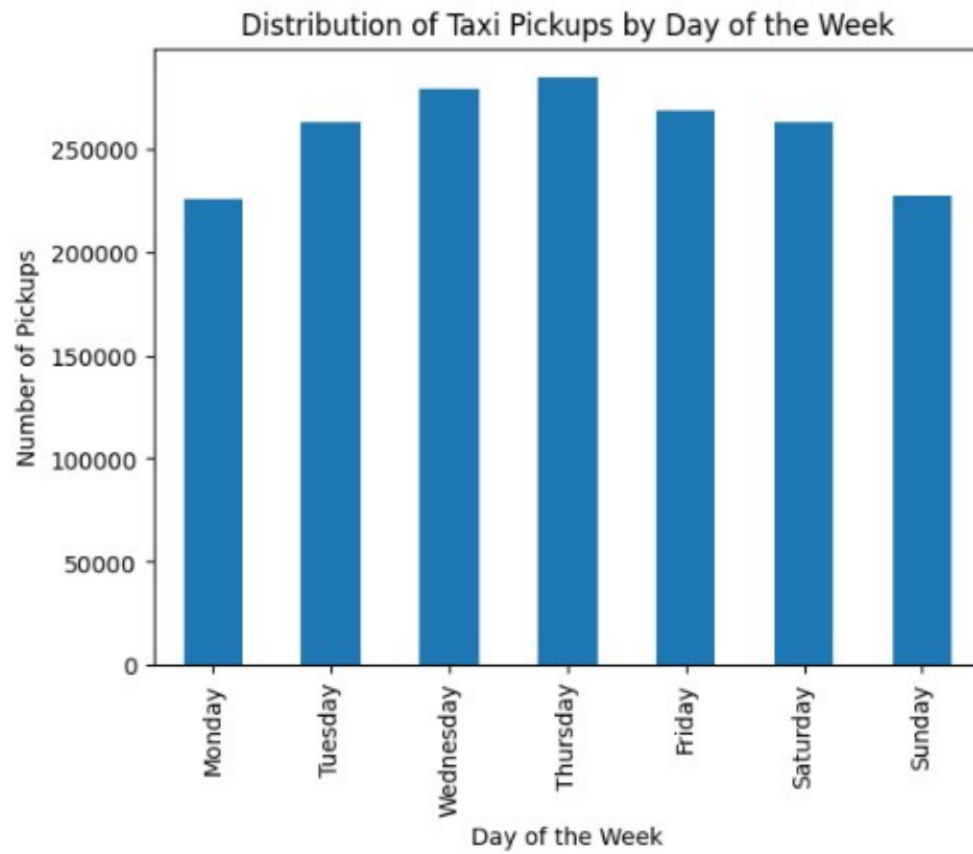
Overall, the EDA demonstrates that taxi demand is strongly influenced by time, day and location. A data-driven operating strategy should therefore dynamically position vehicles in high-demand zones, increase supply ahead of peak hours, optimise routes with persistent low speeds, and Use the observed time-, distance- and vendor-level fare patterns to inform pricing decisions while maintaining competitive rates and customer value.

Reports/Graphs

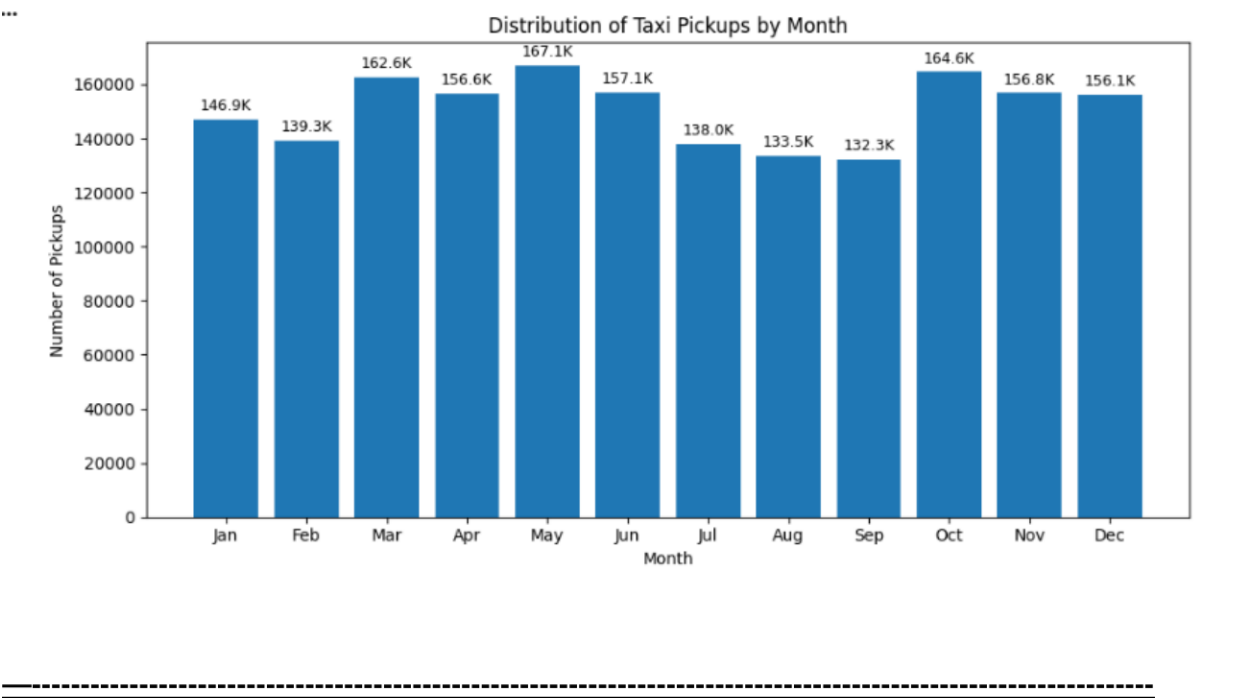
Find and show the hourly trends in taxi pickups



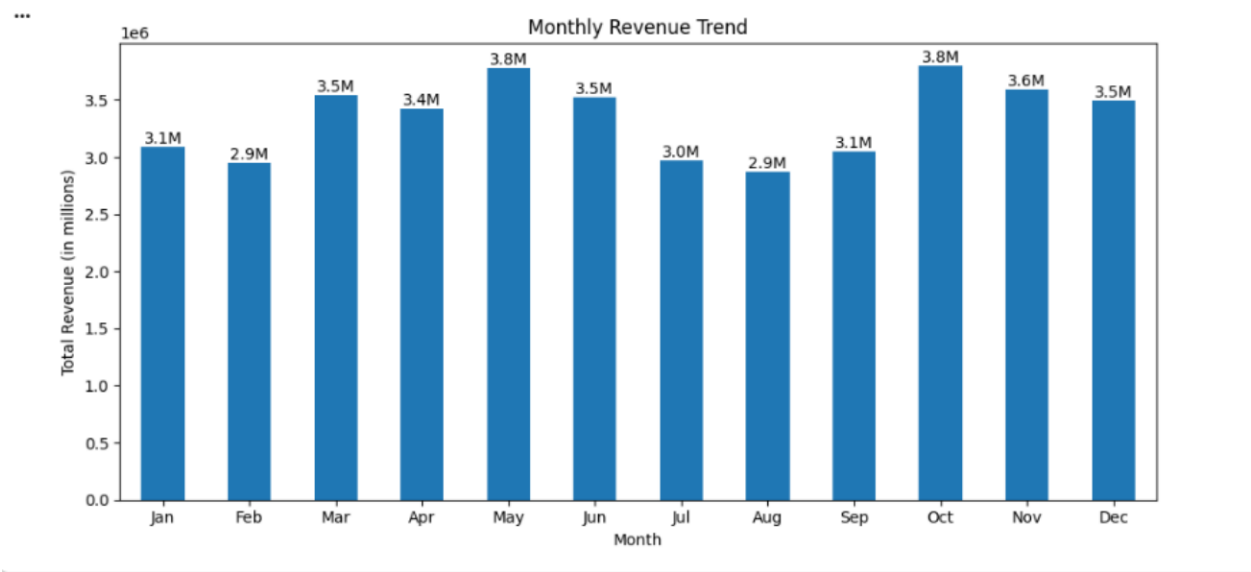
Find and show the daily trends in taxi pickups (days of the week)



Show the monthly trends in pickups

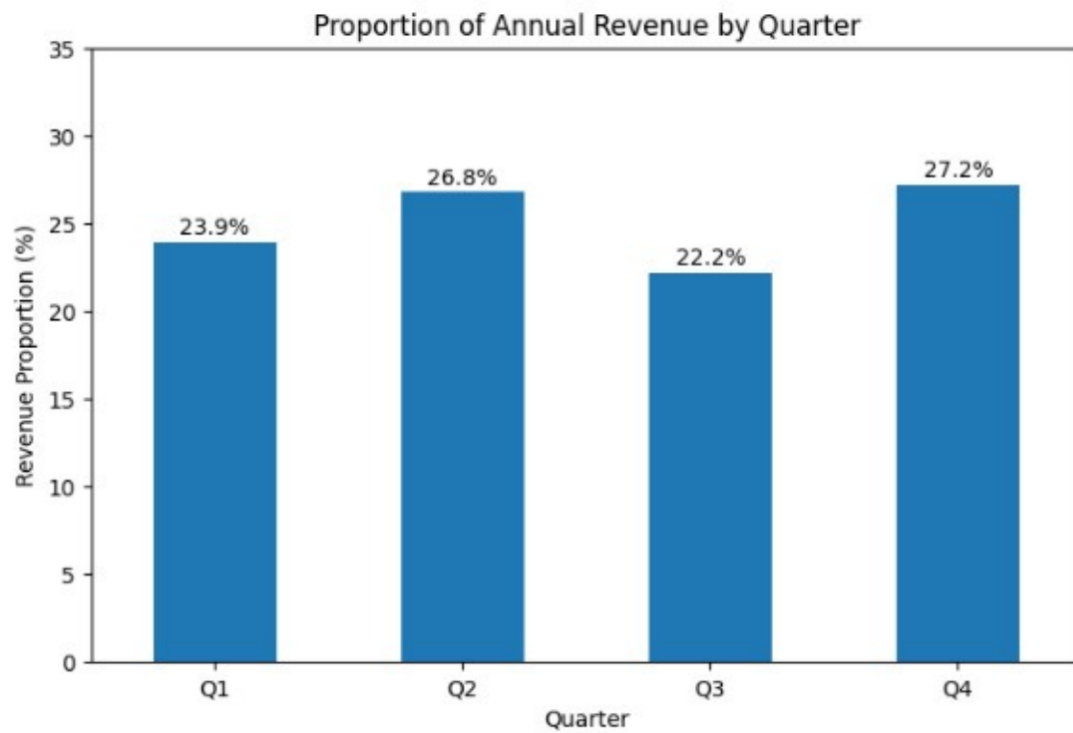


Group data by month and analyse monthly revenue



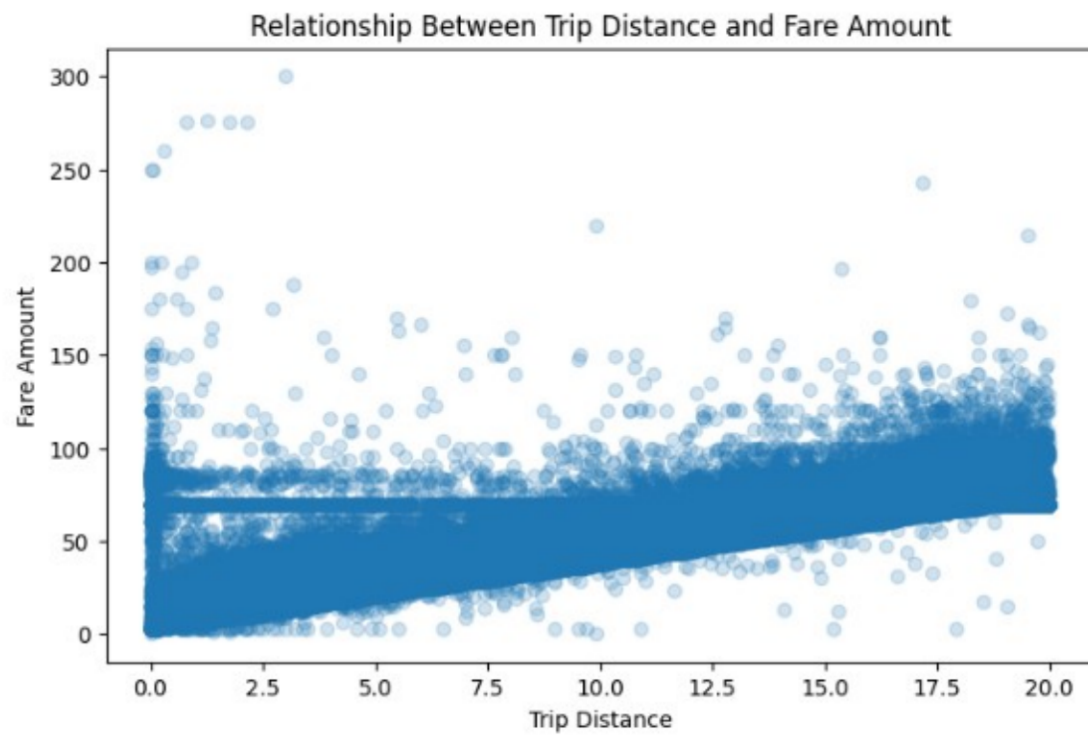
Show the proportion of each quarter of the year in the revenue

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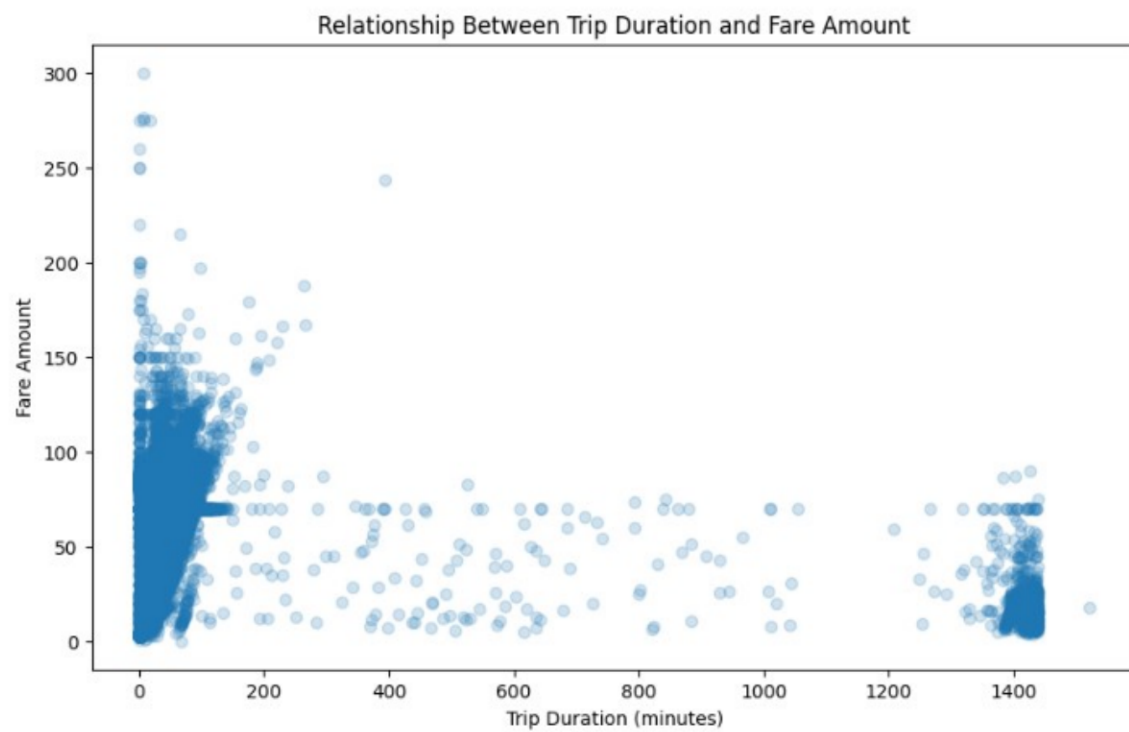
Show how trip fare is affected by distance

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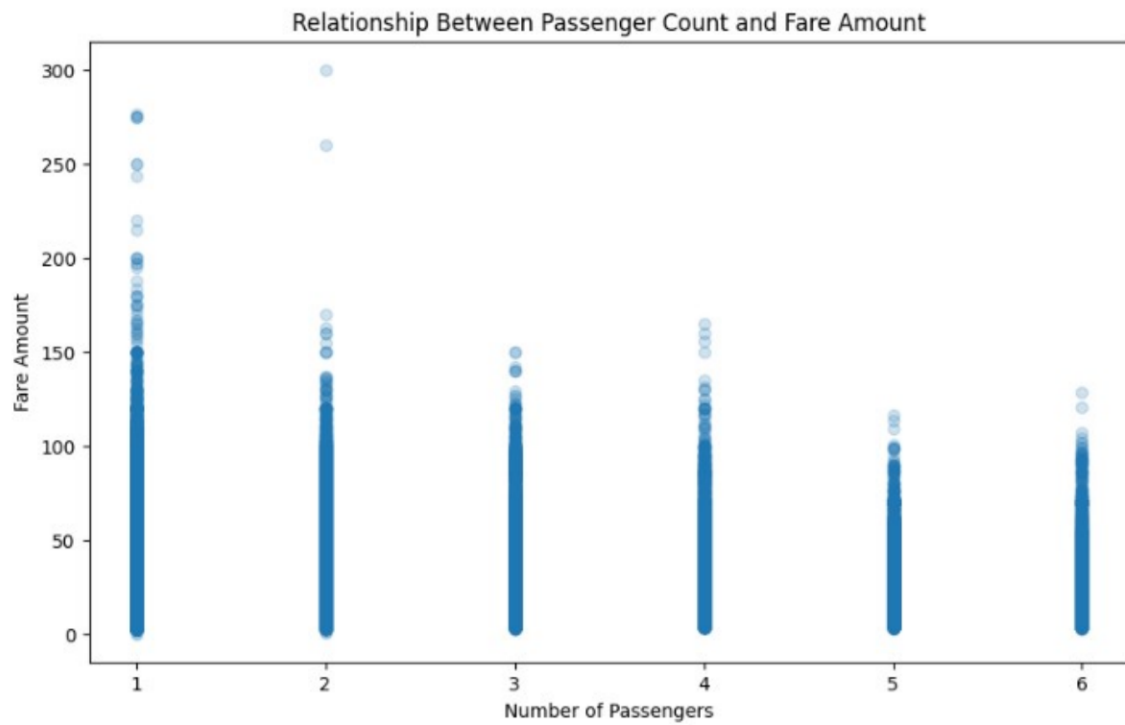
Show relationship between fare and trip duration

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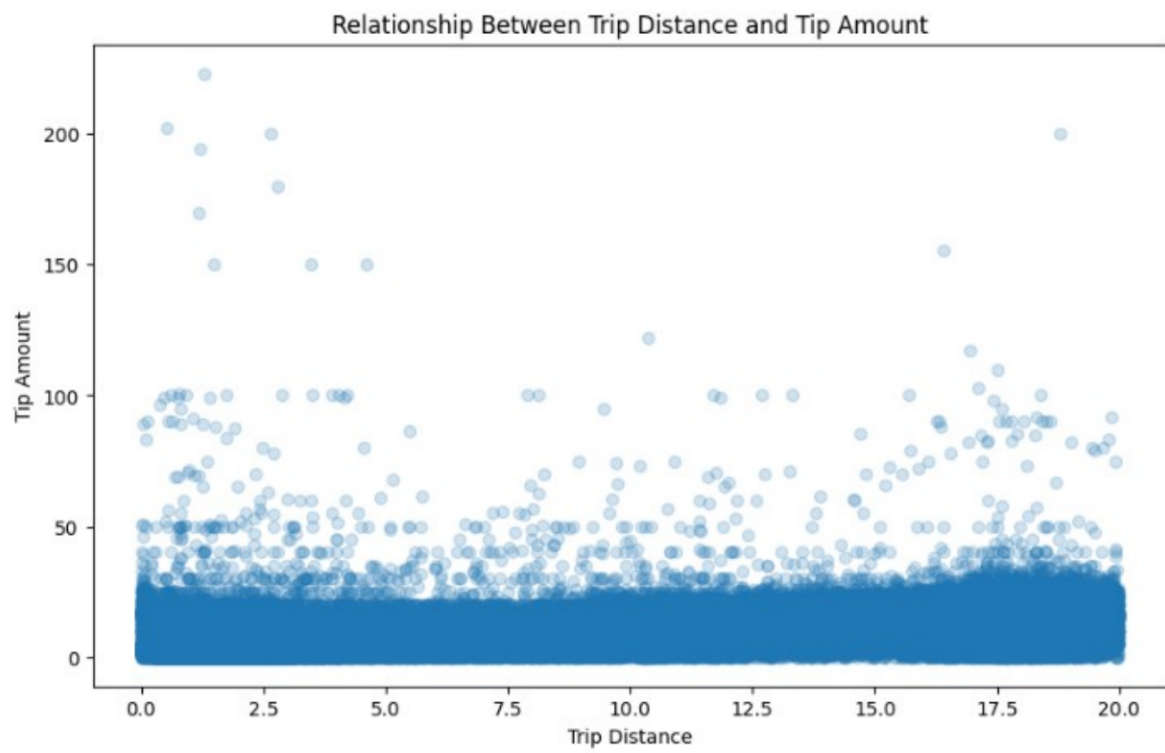


Show relationship between fare and number of passengers

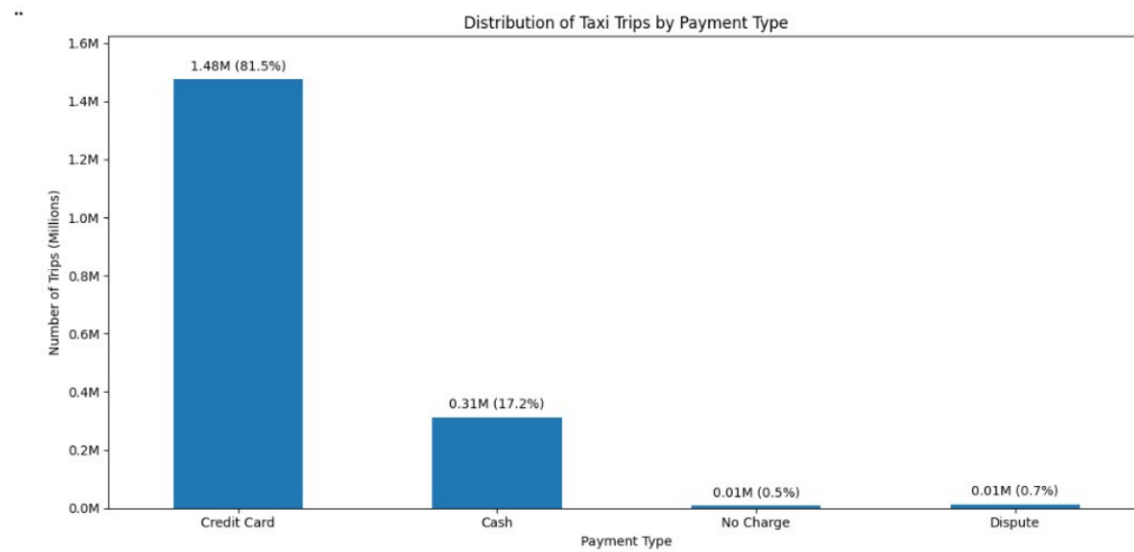
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Show relationship between tip amount and trip distance



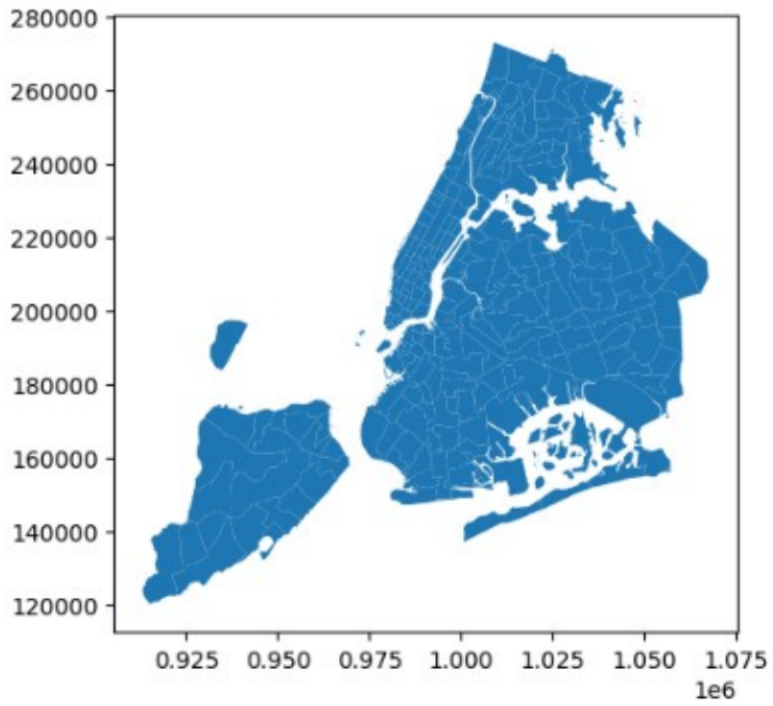
Analyse the distribution of different payment types (payment_type)



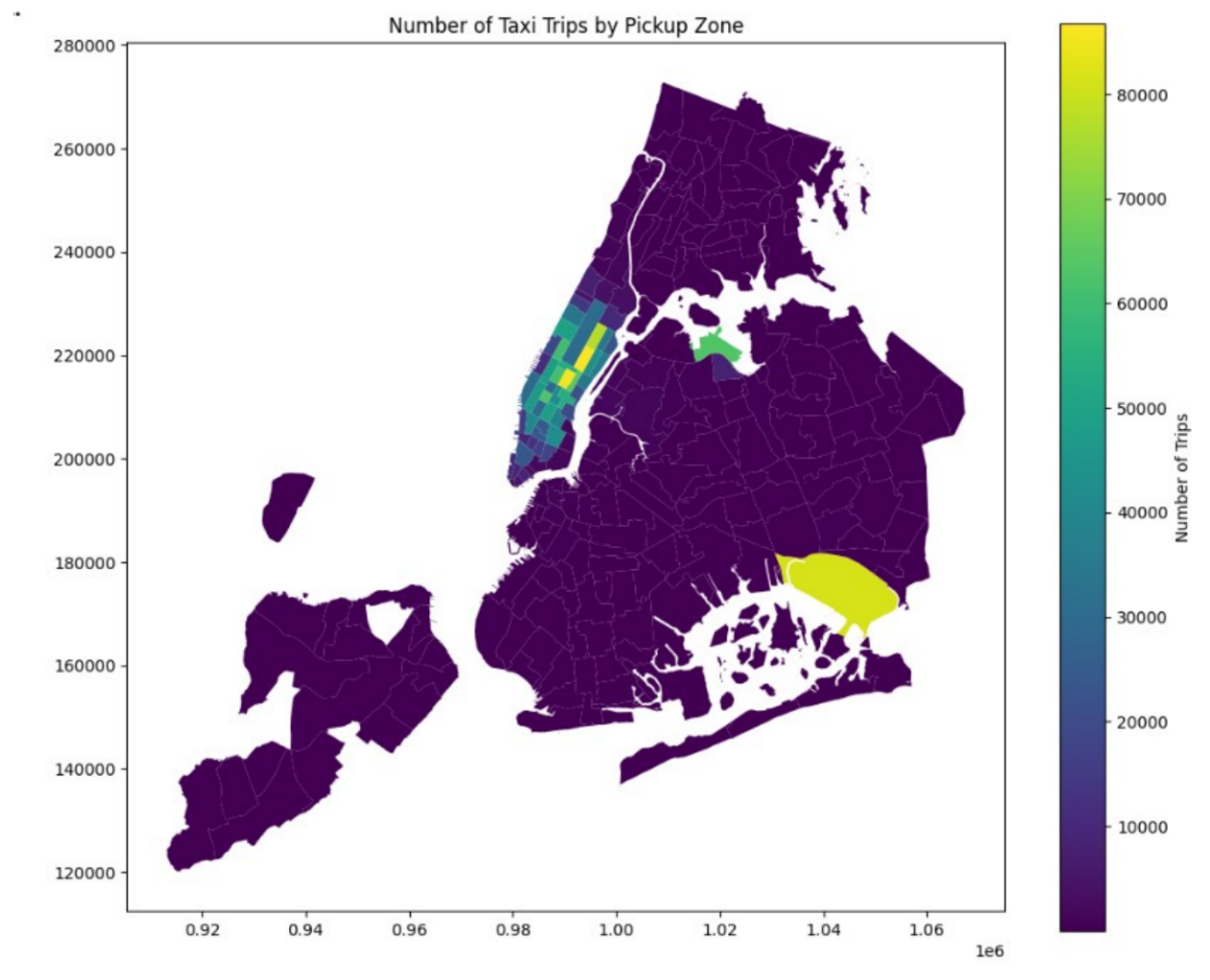
Plotting Zones

none

<Axes: >

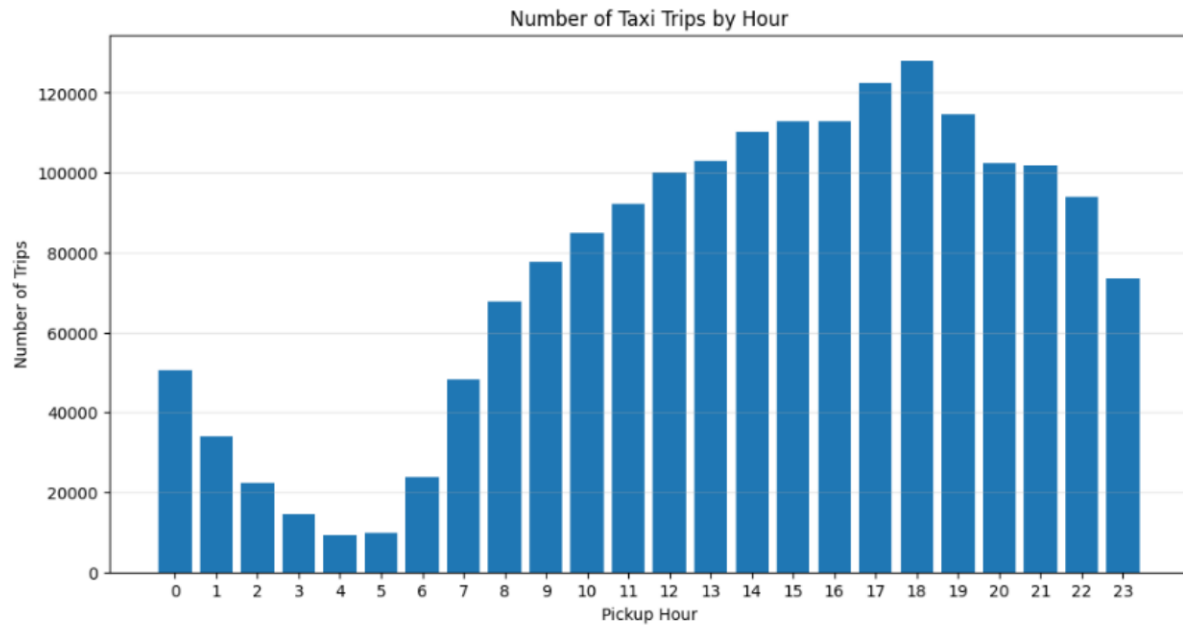


Plot a color-coded map showing zone-wise trips



Visualise the number of trips per hour and find the busiest hour

Busiest hour: 18
Number of trips: 128061



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Hourly drop-off trends for the top 10 drop-off zones

